L1 and L2 Reading of Mental State Inferences: Evidence from Sentence Rating and Eyetracking

Gricean models of language comprehension posit that pragmatic intent is computed after word meaning; however, other work indicates that social information, like speaker identity, is processed as early as semantic information[1]. Here, we assess the automaticity of social inferences during reading by comparing mental state and logical inferences. Mental states reflect the inner thoughts, feelings, goals, and beliefs of others[2], and this "mindreading" capacity depends on the availability of cognitive resources[3]. We examine this in one interesting population, bilinguals, who regularly experience fluctuations in cognitive load and social constraints based on language context. For instance, some evidence suggests that L2 reading involves greater processing demands than L1 reading[4]. Other evidence indicates processing of emotional information, one aspect of mental states, in the L2 is more efficient than the L1[5], and bilinguals may be more attentive to social cues in their L2 than L1[6]. Thus, using a sentence rating paradigm (E1) and natural reading with eyetracking (E2), we test how quickly L1/L2 readers comprehend mental state inferences compared to logical inferences.

The stimuli consisted of 138 original English sentence-pairs in three inference type conditions: logical, mental state, incoherent (Table 1). Sentence 1 introduced a human target in a context (unique to each condition). Sentence 2 featured the same target completing an action (identical across all conditions). In the logical condition, coherence between the context and action required a deductive inference based on world knowledge, whereas in the mental state condition, coherence was achieved through an inference of the target's internal mental state. An incoherent condition (context unrelated to the action) was included as baseline.

In E1 (norming), 61 English/French bilingual adults rated the sentences on coherence and mentalizing required for comprehension. Mental state inferences were rated higher in mentalizing than logical inferences ($p < .001$), and incoherent trials were rated lowest on both questions. Critically, despite no difference in L1 vs. L2 rating for linguistic coherence, L2 readers were significantly more likely than L1 readers to over-attribute mentalizing to logical inferences, suggesting increased sensitivity to social cues in the L2 ($p < .001$; Fig 1).

In E2, a new sample of 57 English/French bilingual adults read the same sentences while their eye movements were recorded (Table 2). We extracted gaze duration, total reading time, and regressions-out[7] across the full action sentence, which was identical across conditions. Early reading time (gaze duration) was not different between logical vs. mental state items, nor between L1 vs. L2 readers. Late reading times were significantly higher for mental state inferences than logical inferences, but only among L2 readers (total reading time $p = .04$; regressions-out $p < .001$; Fig 2).

In sum, although mental state and logical inferences were rated as conceptually distinct, mental state inferences, irrespective of L1/L2, did not slow down early reading times compared to logical inferences. This finding indicates that integration of mental state information is an automatic component of social text comprehension. Despite slower reading times during later stages of reading, L2 readers were more sensitive than L1 readers to mental states in their ratings, even in cases where mentalizing was not necessary for comprehension. This suggests that bilinguals in L2 contexts are more attentive to social cognitive cues in the environment.

Table 1: *Sample stimuli used across both experiments*

	LOGICAL inference	MENTAL STATE inference	Incoherent
Sentence 1 (Context)	Jane took out the house keys	Jane read about the increase in crime	Jane had a fancy pencil case
Sentence 2 (Action)	She locked the front door that day		

Figure 1: *E1 interaction results*

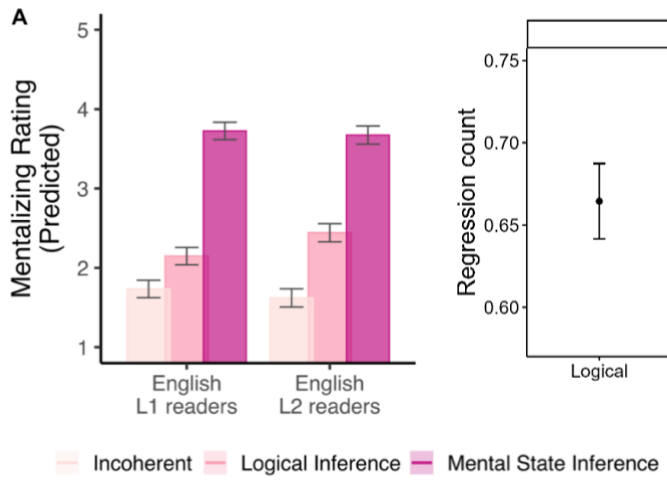

Figure 2: *E2 interaction results (Regressions Out)*

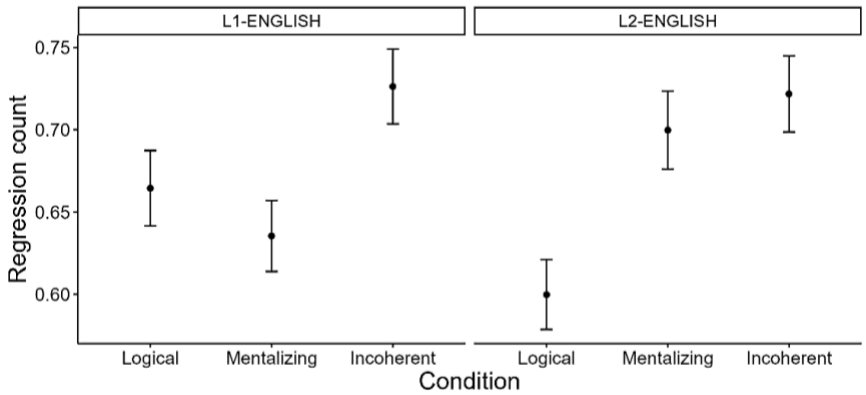

Table 2*: Similar language profiles across participants in Experiment 1 and Experiment 2*

	English L1		English L2	
	E1 (N=31)	*E2 (N=28)*	*E1 (N=30)*	*E2 (N=29)*
Mean % English Use (SD)	83.1 (13.3)	76.2 (15.7)	53.3 (22.4)	47.35 (24.42)
Mean % French Use (SD)	16.9 (13.3)	22.7 (15.5)	46.7 (22.4)	49.5 (24.77)

[1] Van Berkum, J. J., Van den Brink, D., Tesink, C. M., Kos, M., & Hagoort, P. (2008). The neural integration of speaker and message. *Journal of cognitive neuroscience*, *20*(4), 580-591.
[2] Harris, L. T. (2017). *Invisible mind: Flexible social cognition and dehumanization*. MIT Press.
[3] Maehara, Y., & Saito, S. (2013). Cognitive load on working memory both encourages and discourages reasoning bias regarding the mental states of others. *Australian Journal of Psychology*, *65*(3), 163-171.
[4] Bromberek-Dyzman, K., & Rataj, K. (2016). Irony comprehension in the nonnative language comes at a cost. *Psychology of Language and Communication*, *20*(3), 336.
[5] Keysar, B., Hayakawa, S. L., & An, S. G. (2012). The foreign-language effect: Thinking in a foreign tongue reduces decision biases. *Psychological science*, *23*(6), 661-668.
[6] Mousley, V. L., MacSweeney, M., & Mercure, E. (2023). Bilingual toddlers show increased attention capture by static faces compared to monolinguals. *Bilingualism: Language and Cognition*, 26(4).
[7] Rayner, K. (1998). Eye movements in reading and information processing: 20 years of research. *Psychological bulletin*, *124*(3), 372.